

Contraction Theory for Learning-based Control of Nonlinear Dynamical Systems—Part 3: Conformal Inference

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- In the following, $\|\cdot\|$ denotes the Euclidean norm for vectors and the induced 2-norm for matrices, i.e., $\|\cdot\| = \|\cdot\|_2$, unless otherwise noted.
- For a symmetric square matrix $S \in \mathbb{R}^{n \times n}$, we use the notation $S \succ 0$, $S \succeq 0$, $S \prec 0$, and $S \preceq 0$ for the positive definite, positive semi-definite, negative definite, and negative semi-definite matrices, respectively.
- This lecture is based on [1, 2].

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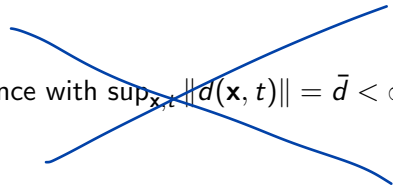
References

What is missing?

Let's again consider the following perturbed nonlinear dynamical system:

$$\dot{\mathbf{x}} = f(\mathbf{x}, t) + d(\mathbf{x}, t)$$

where $d(\mathbf{x}, t)$ represents external disturbance with $\sup_{\mathbf{x}, t} \|d(\mathbf{x}, t)\| = \bar{d} < \infty$.



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What is unrealistic here?

How do we know that $d(\mathbf{x}, t)$ is bounded in the first place?

We could sample data of $d(\mathbf{x}, t)$ to empirically check if it is bounded, but do you think this is sufficient in safety-critical applications, e.g., in space?

$$q = \phi(p)$$

This issue represents a central difficulty in machine learning.

Suppose a Llama large language model (LLM) outputs the following prompt:

=====

Test prompt: What should I do with my life?

Research topic: aerospace robotics

=====

Generating text before fine-tuning

=====

What should I do with my life? This is a question that has plagued many people for years. It's a question that can be both exciting and terrifying. First, let me tell you that there is no one-size-fits-all answer. What works for one person may not work for another. What's most important is to take the time to reflect on your values, passions, and strengths. Ask yourself:

- * What do I love doing?
- * What makes me feel fulfilled?
- * What ...

$$q_i = \phi(p_i), i=1, \dots, N$$

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How can we verify that the LLM does not hallucinate, meaning its output is a valid and accurate response to the input prompt?

Suppose that we have a predictor for our system's uncertainty with N real-world data points describing the uncertainty.

Can we mathematically say anything about the uncertainty **without knowing anything about the prediction structure and data distribution?**



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Definition (Statistic)

A statistic is any measurable function that depends only on the observed sampled data $(S_1, S_2, \dots, S_N)$

$$y = T(S_1, S_2, \dots, S_N).$$

For example,

~~$$T_{\text{ord}}(S_1, \dots, S_N) = (S_{\pi(1)}, \dots, S_{\pi(N)})$$~~

~~are referred to as the order statistics.~~

$$N \text{ data} : (S_1, \dots, S_N)$$

$$E_S[\bar{S}] = \sum_{i=1}^N \frac{1}{N} S_i$$

$$E[S] = ?$$

$$= \int S d\mu$$

The goal of statistical inference is to learn characteristics of a population from statistics of data samples.

This implies that statistical inference naturally encompasses all the (parametric) estimation approaches.

The term “estimation” typically refers to the process of determining unknown parameters or other quantities of interest that characterize the population.

The key difficulty in statistical inference is that **the characteristic parameters of the population are themselves unknown** because, at the beginning, everything is unknown.

Examples include, but are not limited to

- Probably approximately correct (PAC)-Bayes [3]
- Scenario approach [4, 5]
- Conformal inference [1, 2]

We focus on conformal inference, which may be most compatible with nonlinear control/estimation **without any structural and distributional priors**¹².

¹Ting-Wei Hsu and Hiroyasu Tsukamoto. “Statistical guarantees in data-driven nonlinear control: Conformal robustness for stability and safety”. In: *IEEE Control Systems Letters* 9 (2025), pp. 997–1002. DOI: 10.1109/LCSYS.2025.3578062.

²Sihang Wei, Melkior Ornik, and Hiroyasu Tsukamoto. “Conformal contraction for robust nonlinear control with distribution-free uncertainty quantification”. In: *IEEE Conference on Decision and Control*, to appear. Dec. 2025. DOI: TBA.

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$$S_1, \dots, S_N \sim \mathcal{P}(S_1, \dots, S_N)$$

Let $S_1, \dots, S_N \in \mathbb{R}$ be real-valued random variables.

Definition (Exchangeability)

For any permutation $\sigma \in \mathcal{S}_N$,

$$S_{\sigma(1)}, \dots, S_{\sigma(N)} \sim \tilde{\mathcal{D}}(S_{\sigma(1)}, \dots, S_{\sigma(N)})$$

$$(S_1, \dots, S_N) \stackrel{d}{=} (S_{\sigma(1)}, \dots, S_{\sigma(N)})$$

$$\mathcal{D} = \tilde{\mathcal{D}}$$

where $\mathcal{S}_N$ is the set containing all the permutations, $\stackrel{d}{=}$ means that the two vectors have the same distribution. In particular, we have

$$\mathbb{P}[(S_1, \dots, S_N) \in A] = \mathbb{P}[(S_{\sigma(1)}, \dots, S_{\sigma(N)}) \in A]$$

for all ³measurable sets A .

³the sets where the probability $\mathbb{P}[\cdot \in A]$ is well-defined

$$(S_1, S_2, S_3) \sim \mathcal{P}$$

$$(S_2, S_3, S_1) \sim \mathcal{P}$$

$$6(1) = 2$$

$$6(2) = 3$$

$$6(3) = 1$$

3! permutations

(1, 2, 3)

(1, 3, 2)

⋮

iid $\Rightarrow$ exchangeable
 $\Leftarrow$

$$\mathbb{P}[(S_1, \dots, S_N) \in A] = \mathbb{P}[(S_{6(1)}, \dots, S_{6(N)}) \in A]$$

Suppose that

- The vector $(S_1, \dots, S_N)$ is exchangeable and
- The S_i are almost surely distinct, i.e., $\mathbb{P}[S_i \neq S_j] = 1$ for $i \neq j$

and define the rank vector as

$$\text{Rank}(S_1, \dots, S_N) = \pi \in \mathcal{S}_N$$

set of permutation

such that

$$S_{\pi(1)} \leq S_{\pi(2)} \leq \dots \leq S_{\pi(N)}.$$

$$S_1 = 5$$

$$S_2 = 4$$

$$S_3 = 7$$

$$S_2 < S_1 < S_3$$

$$\pi(1) \quad \pi(2) \quad \pi(3)$$

$$\pi(1) = 2$$

$$\pi(2) = 1$$

$$\pi(3) = 3$$

$$\text{index} = \pi(\text{rank})$$

$$\text{rank} = \pi^{-1}(\text{index})$$

Definition (Rank Statistics)

A statistic T of observed samples $(S_1, \dots, S_N)$ defined only by their rank vector is called a rank statistic.

Theorem 1

Suppose that

- *The random vector $(S_1, \dots, S_N)$ is exchangeable and*
- *The S_i are almost surely distinct, i.e., $\mathbb{P}[S_i \neq S_j] = 1$ for $i \neq j$.*

Then we have

$$\mathbb{P}[\text{Rank}(S_1, \dots, S_N) = \pi] = \frac{1}{N!} \text{ for all } \pi \in \mathcal{S}_N.$$

(Handwritten blue wavy line under the fraction denominator and $N!$ below it)

Proof: Let A_π be a set of $(S_1, \dots, S_N)$ defined as

$$A_\pi = \{(S_1, \dots, S_N) : S_{\pi(1)} < \dots < S_{\pi(N)}\}$$

where $\pi \in \mathcal{S}_N$. Since we have for any permutation $\sigma \in \mathcal{S}_N$

$$(S_1, \dots, S_N) \stackrel{d}{=} (S_{\sigma(1)}, \dots, S_{\sigma(N)})$$

due to exchangeability, we get

$$\mathbb{P}[(S_1, \dots, S_N) \in A_\pi] = \mathbb{P}[(S_{\sigma(1)}, \dots, S_{\sigma(N)}) \in A_\pi]$$

by def. of
rank vector

Take $\pi \in \mathcal{P}_N$

Example $\pi(1) = 2$ $S_1 = 6$
 $\pi(2) = 3$ $S_2 = 5$ is $(S_1, S_2, S_3) \in A_\pi$?
 $\pi(3) = 1$ $S_3 = 7$

$$A_\pi = \left\{ (S_1, S_2, S_3) \text{ s.t. } \begin{array}{ccc} S_{\pi(1)} & < & S_{\pi(2)} < S_{\pi(3)} \\ \parallel & & \parallel & \parallel \\ S_2 & & S_3 & S_1 \end{array} \right\}$$

Also, by the definition of A_π , we have

$$\mathbb{P}[(S_1, \dots, S_N) \in A_\pi] = \mathbb{P}[\text{Rank}(S_1, \dots, S_N) = \pi]$$

$$\mathbb{P}[(S_1, \dots, S_N) \in A_\pi] = \mathbb{P}[\text{Rank}(S_{\sigma(1)}, \dots, S_{\sigma(N)}) = \pi]$$

and thus

$$\mathbb{P}[\text{Rank}(S_1, \dots, S_N) = \pi] = \mathbb{P}[\text{Rank}(S_{\sigma(1)}, \dots, S_{\sigma(N)}) = \pi].$$

Finally, for any $\pi, \varpi \in \mathcal{S}_N$, let us define σ as

$$\sigma = \pi^{-1} \circ \varpi \Rightarrow \varpi = \pi \circ \sigma. \quad (\text{property of permutation})$$

Then we have

$$\begin{aligned} \text{Rank}(S_{\sigma(1)}, \dots, S_{\sigma(N)}) = \pi &\Leftrightarrow S_{\pi(\sigma(1))} < \dots < S_{\pi(\sigma(N))} \\ &\Leftrightarrow \text{Rank}(S_1, \dots, S_N) = \varpi \end{aligned}$$

which yields

$$\mathbb{P}[\text{Rank}(S_1, \dots, S_N) = \pi] = \mathbb{P}[\text{Rank}(S_1, \dots, S_N) = \varpi]$$

for any $\pi, \varpi \in \mathcal{S}_N$. Using this relation with $|\mathcal{S}_N| = 1/n!$ completes the proof.

$$\begin{array}{ccc} & \pi & \\ 1 & \leftrightarrow & 3 \\ 2 & \leftrightarrow & 2 \\ 3 & \leftrightarrow & 1 \\ & \pi^{-1} & \\ \Rightarrow & \pi \circ \pi^{-1} & = \mathbb{I} \end{array}$$

Corollary 2

Suppose again that

- *The random vector $(S_1, \dots, S_N)$ is exchangeable and*
- *The S_i are almost surely distinct, i.e., $\mathbb{P}[S_i \neq S_j] = 1$ for $i \neq j$.*

Then we have

$$\mathbb{P}[R_i = r] = \frac{1}{n} \text{ for all } r = 1, \dots, N$$

where $R_i = \pi(i)$ is the i -th element of $\text{Rank}(S_1, \dots, S_N) = \pi \in \mathcal{S}_N$.

Proof: $|\mathcal{S}_N| = n!$ and we have $(n-1)!$ permutation options if we fix one of $(S_1, \dots, S_N)$.

Theorem 3 (Conformal Inference)

Suppose that

- *The random vector $(S_1, \dots, S_N, S_{N+1})$ is exchangeable and*
- *The S_i are almost surely distinct, i.e., $\mathbb{P}[S_i \neq S_j] = 1$ for $i \neq j$.*

Let $\pi = \text{Rank}(S_1, \dots, S_N)$. Then we have

$$\mathbb{P} \left[S_{N+1} \leq S_{\pi(\lceil (1-\delta)(N+1) \rceil)} \right] \in \left[1 - \delta, 1 - \delta + \frac{1}{N+1} \right]$$

where $\delta \in (0, 1)$ and $\lceil \cdot \rceil$ is the ceiling function.

Proof: We have

$$\begin{aligned}\mathbb{P} [S_{N+1} \leq S_{\pi(\lceil(1-\delta)(N+1)\rceil)}] &= \mathbb{P} [R_{N+1} \leq \lceil(1-\delta)(N+1)\rceil] \\ &= \mathbb{P} \left[\bigcup_{i=1}^{\lceil(1-\delta)(N+1)\rceil} \{R_{N+1} = i\} \right] = \sum_{i=1}^{\lceil(1-\delta)(N+1)\rceil} \mathbb{P} [R_{N+1} = i] = \frac{\lceil(1-\delta)(N+1)\rceil}{N+1}\end{aligned}$$

where R_{N+1} denotes the ranking of S_{N+1} , the third equality is due to the fact that the events $\{R_{N+1} = i\}$ are disjoint, and the last equality is due to Corollary 2.

Then by definition of the ceiling function, we have

$$\frac{\lceil (1 - \delta)(N + 1) \rceil}{N + 1} \geq \frac{(1 - \delta)(N + 1)}{N + 1} = 1 - \delta$$

and

$$\frac{\lceil (1 - \delta)(N + 1) \rceil}{N + 1} \leq \frac{(1 - \delta)(N + 1) + 1}{N + 1} = 1 - \delta + \frac{1}{N + 1}$$

which completes the proof.

The inequality

$$\mathbb{P} \left[S_{N+1} \leq S_{\pi(\lceil (1-\delta)(N+1) \rceil)} \right] \in \left[1 - \delta, 1 - \delta + \frac{1}{N+1} \right]$$

can be interpreted as

$$\mathbb{P} \left[S_{N+1} \leq \lceil (1 - \delta)(N + 1) \rceil\text{-th smallest value in } (S_1, \dots, S_N) \right] \in \left[1 - \delta, 1 - \delta + \frac{1}{N+1} \right].$$

Surprisingly, we did not assume anything about the underlying probability distribution. What's wrong with this bound?

The data you have $\{(x_i, y_i)\}_{i=1}^N$ has to be exchangeable with the unknown, real-world data (x_{N+1}, y_{N+1}) .

The bound assumes randomness in $(S_1, \dots, S_N)$, but in practice, this is fixed ($\rightarrow$ beta distribution, see [here](#)).

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Consider an unknown function $\mathcal{F}$ with inputs $\mathcal{X}$ and outputs $\mathcal{Y}$

$$\mathcal{Y} = \mathcal{F}(\mathcal{X}) \tag{1}$$

and let $\hat{\mathcal{F}}$ be a prediction model of $\mathcal{F}$.

The function $\hat{\mathcal{F}}$ can represent various things.

- Machine learning models predicting real-world phenomena, e.g., stock prices
- Neural networks predicting uncertainty in dynamical systems
- Reinforcement learning agents predicting cumulative rewards
- LLMs/vision language models (VLMs) such as ChatGPT

Importantly, **we will not assume anything about the structure of $\mathcal{F}$ or the distribution of data points.**

Let us define the prediction nonconformity score as

$$s(\mathcal{X}, \mathcal{Y}) = \|\mathcal{Y} - \hat{\mathcal{F}}(\mathcal{X})\|$$

where $\mathcal{Y}$ is the ground truth output given by (1).

Given data samples $\{(\mathcal{X}_i, \mathcal{Y}_i)\}_{i=1}^N$, we quantify the prediction nonconformity scores for the conformal inference as

$$S_i = s(\mathcal{X}_i, \mathcal{Y}_i) = \|\mathcal{Y}_i - \hat{\mathcal{F}}(\mathcal{X}_i)\|, \quad i = 1, \dots, N.$$

Also, let

$$S_{\text{new}} = s(\mathcal{X}_{\text{new}}, \mathcal{Y}_{\text{new}}) = \left\| \mathcal{Y}_{\text{new}} - \hat{\mathcal{F}}(\mathcal{X}_{\text{new}}) \right\|$$

represent the nonconformity score of unseen, future data sample $(\mathcal{X}_{\text{new}}, \mathcal{Y}_{\text{new}})$.

Suppose that

- The random vector $(S_1, \dots, S_N, S_{\text{new}})$ is exchangeable and
- The scores are almost surely distinct.

Let $\pi = \text{Rank}(S_1, \dots, S_N)$. Then Theorem 3 implies that

$$\mathbb{P} \left[S_{\text{new}} \leq S_{\pi(\lceil (1-\delta)(N+1) \rceil)} \right] \in \left[1 - \delta, 1 - \delta + \frac{1}{N+1} \right].$$

More intuitively, we have

$$\mathbb{P} \left[\left\| \mathcal{Y}_{\text{new}} - \hat{\mathcal{F}}(\mathcal{X}_{\text{new}}) \right\| \leq \lceil (1 - \delta)(N + 1) \rceil\text{-th smallest in } (S_1, \dots, S_N) \right] \in \left[1 - \delta, 1 - \delta + \frac{1}{N + 1} \right].$$

We often just use the lower bound to get

$$\mathbb{P} \left[\left\| \mathcal{Y}_{\text{new}} - \hat{\mathcal{F}}(\mathcal{X}_{\text{new}}) \right\| \leq \lceil (1 - \delta)(N + 1) \rceil\text{-th smallest in } (S_1, \dots, S_N) \right] \geq 1 - \delta.$$

In words, this means the unseen prediction error $\left\| \mathcal{Y}_{\text{new}} - \hat{\mathcal{F}}(\mathcal{X}_{\text{new}}) \right\|$ is bounded by $\lceil (1 - \delta)(N + 1) \rceil$ -th smallest value in $(S_1, \dots, S_N)$ at least with probability $1 - \delta$.

This bound is no longer empirical but statistically rigorous. How can we use it with contraction theory?

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